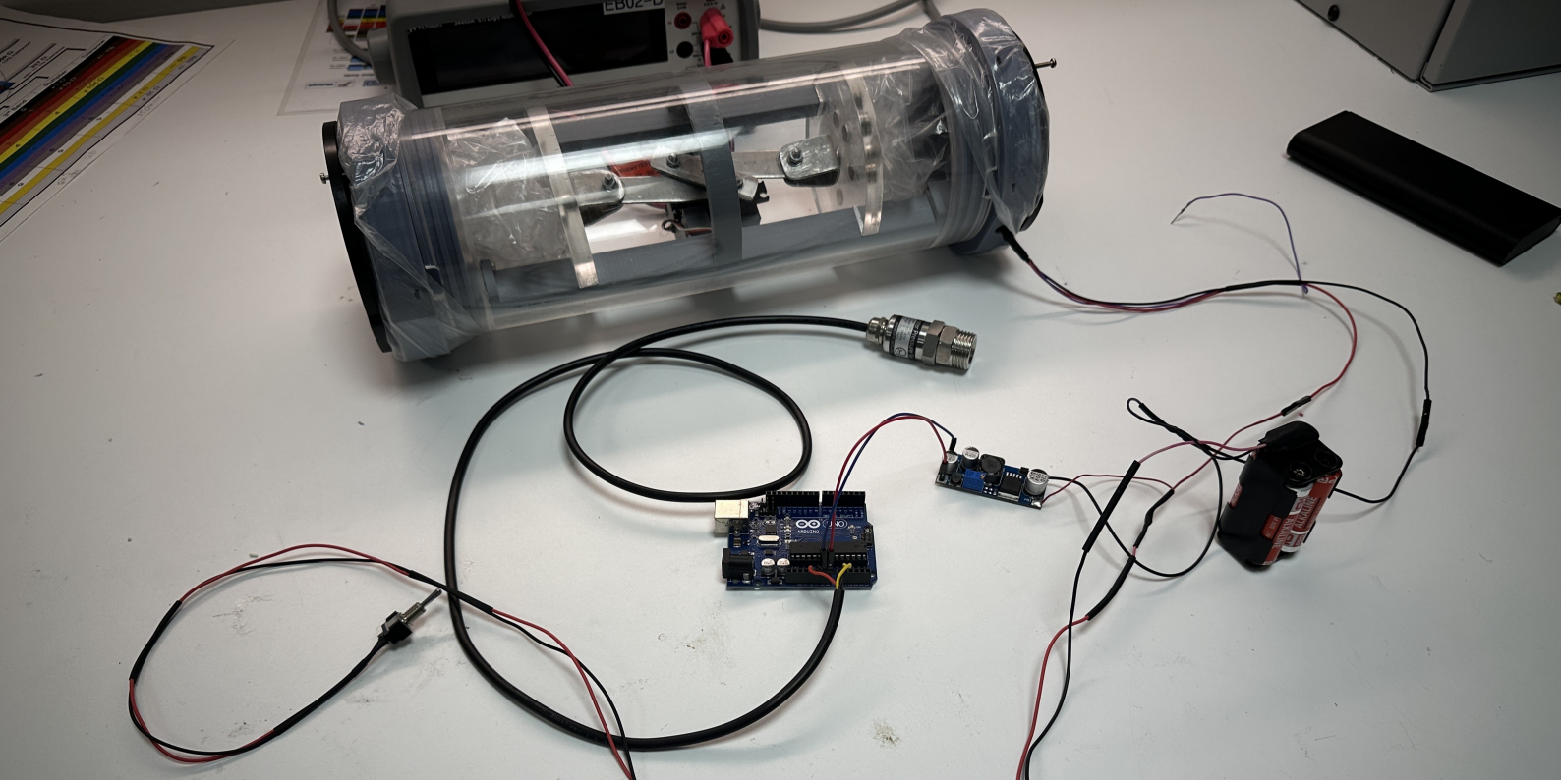


Buyoancy Engine

Development of a simple buoyancy engine

Mate Papp, Milos Jozek, Ricardo Avila, Tuguldur Baatarkhuu

Applied Industrial Electronics, AIE1-4, 2026-10

Semester Project



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Applied Industrial Electronics. AIE1-4, 2025-12

Semester Project



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AI was used in this report mainly for spellchecking and formatting with tools like Grammarly and LanguageTool. Additionally, generative AI tools, namely Microsoft Copilot, ChatGPT, and Google Gemini, were used for learning certain concepts in the field of physics, for help in coding in Arduino IDE, and for assistance while writing the report in Overleaf (for certain commands and error codes).



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Abstract:

This study is relevant for academic institutions as it aims to create a smaller and simpler buoyancy engine that can demonstrate diving and surfacing movement, seeing as current buoyancy engines are expensive and overly complex for small-scale applications. This buoyancy engine works based on Archimedes' principle, essentially changing its density relative to the surrounding water, where water is taken in and out of a bladder/ballast inside the buoyant device. In comparison to existing buoyancy engines and other methods of vertically moving an underwater vehicle, this one requires far less power and energy to work, whereby its cost and size are minimized.
Although not as reliable for environmental or scientific research, the presented buoyancy engine contributes well to the learning process for students and learners

The content of this report is freely available, but publication (with reference) may only be pursued due to agreement with the author.

Contents

Preface **v**

1 Introduction **1**
1.1 Problem Analysis . . . 3
1.1.1 Economical . . . 3
1.1.2 Path Planning . . . 4
1.1.3 Energy Consumption . . . 4
1.1.4 Environmental Impact . . . 4
1.2 Problem Statement . . . 5

2 Background and theory **6**
2.1 Literature review . . . 6
2.1.1 Existing solutions . . . 6
2.1.1.1 Strengths and weaknesses of the solutions . . . 9
2.2 Theoretical principles . . . 9
2.2.1 Archimedes' principle . . . 9
2.2.2 Density and Buoyancy balance . . . 10
2.2.3 Hydrostatic pressure and depth dependency . . . 11
2.2.4 Energy considerations in volume modulation . . . 11
2.2.5 Integrated buoyancy control . . . 12

3 Design and methodology **13**
3.1 Project Scope and Limitations . . . 13
3.1.1 Project scope: . . . 13
3.1.2 Project limitations: . . . 14
3.2 System Requirements . . . 15
3.3 Design considerations . . . 16
3.3.1 Mass and Buoyancy . . . 16
3.3.2 Sealing . . . 18
3.4 Design . . . 18
3.4.1 Initial Design . . . 19

3.4.2 Final Design . . . 19
3.5 Components selection and list . . . 19
3.5.1 Components Listed . . . 28

4 Implementation and evaluation **29**
4.1 Assembly and integration . . . 29
4.1.1 Hardware . . . 30
4.1.2 Software . . . 30
4.1.2.1 System Overview . . . 31
4.1.2.2 Sensor reading and depth calculation . . . 31
4.1.2.3 Control logic . . . 32
4.1.2.4 Servo Actuation . . . 32
4.2 Testing procedures . . . 35
4.3 Results and Discussion . . . 36

5 Conclusion **38**

Bibliography **39**

A Appendix **42**
.1 3D model of Acrylic tube . . . 42
.2 Arduino UNO datasheet . . . 42
.3 SEN0257 Water pressure sensor datasheet . . . 42
.4 Parallax Servo datasheet . . . 42
.5 Project Repository . . . 42

Preface

This report was prepared as part of the Applied Industrial Electronics course at Aalborg University during the fall semester of 2025. The project focuses on the design, implementation, and testing of a buoyancy engine prototype, integrating mechanical, electrical, and software components.

The report is intended for instructors and students within the field of engineering and documents both the development process and the final system performance.

We would like to thank Fredrik Fogh Sørensen for guidance and feedback throughout the project, as well as Mads Valentin Bram for valuable input and clarifications, and Jeppe Nørgaard Villadsen, the lab technician, for providing access to equipment and facilities.

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Chapter 1

Introduction

The ocean influences the global climate strongly, absorbing about 30% of the carbon dioxide emitted by humans. The oceanic heat capacity is also about 1000 times greater than the atmospheric heat capacity, and it has absorbed more than 90% of the heat stored in the entire planet since 1971 (Argo Program). Slight changes have been observed, even in deep waters, where the environment is typically considered relatively stable. These small changes have a significant impact on the global climate and on the ocean's fauna, flora, and overall environment (Pallayil et al.). As such, the observation and analysis of deep-sea water is of immense importance to devise the appropriate measures. Autonomous profiling floats and other underwater vehicles have come into use for this purpose.

Underwater vehicles are becoming increasingly important for scientific and environmental research, and are widely used in industrial applications, sub-aquatic wildlife, and nature explorations. One example is the Argo program, which has around 4000 active floats deployed around the globe (Argo Program). It is therefore a fact that such submarines and gliders are required to descend and ascend precisely and reliably underwater. For this depth control, the most commonly used technology in gliders is propellers. This method is inefficient since it consumes a significant amount of energy to counterbalance buoyant forces, meaning the propellers must remain in motion even if the desired depth is reached. Hence, underwater expeditions that are several days or weeks long force engineers to focus on energy efficiency while, at the same time, maintaining reliability and accurate performance. This matter leads us to the question of a simpler concept that meets industrial and scientific demands. One of these alternatives is the buoyancy engine. It is based on a principle of altering density by changing volume (Archimedes' principle), either becoming denser than the water surrounding it to descend or less dense to ascend, or becoming heavier, so that the force acting downwards on the object counteracts the upwards buoyant force. Applying this principle, buoyancy engines are outperforming conventional propellers and thrusters (used in typical AUVs) in terms of energy efficiency in vertical movement.

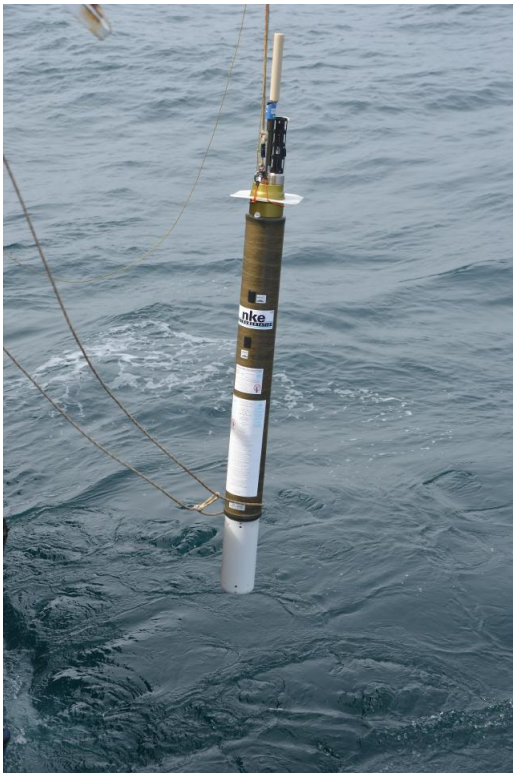


Figure 1: "Deep Arvor Argo float deployment (Ifremer 00619-73078 - 29063)" by Kevin Balem is licensed under CC BY 4.0.

Proven its effectiveness, it lacks the simplicity of propellers, and while in large-scale operations a buoyancy engine-powered setup might be considered a good investment, small-scale operations often prefer thrusters and propellers because of their simplicity and lower cost. Thrusters and electric motors are easily accessible and are widely used for testing and educational purposes, and although buoyancy engines can be built, it is still time- and energy-consuming. Such systems require perfect waterproofing, special materials and components that are not electronics- and robotics-related, and, as a result, it is difficult for student projects. Additionally, internal moving parts and electronics submerged underwater and under pressure mean high risks of reliability and technical difficulties.

In the early 2000s, the Argo (Array for Real-time Geostrophic Oceanography) program was created. It is a collaborative partnership between more than 30 countries that collects information on temperature, salinity, pressure, and biogeochemical components such as oxygen, pH, nitrates, chlorophyll, etc., from inside the ocean using several robotic instruments (floats) that are moved side to side by the ocean currents and move up and down with a buoyancy engine. The core floats go as deep as 2000 meters, and the Deep Argo

Mission floats go all the way down to 6000 meters (**Argo**). Knowing about the existence of such a program is important because it is where these buoyancy engines are applied and used. It is also important to know about this program because Argo floats are not perfect; therefore, they have numerous issues and problems that need to be addressed.

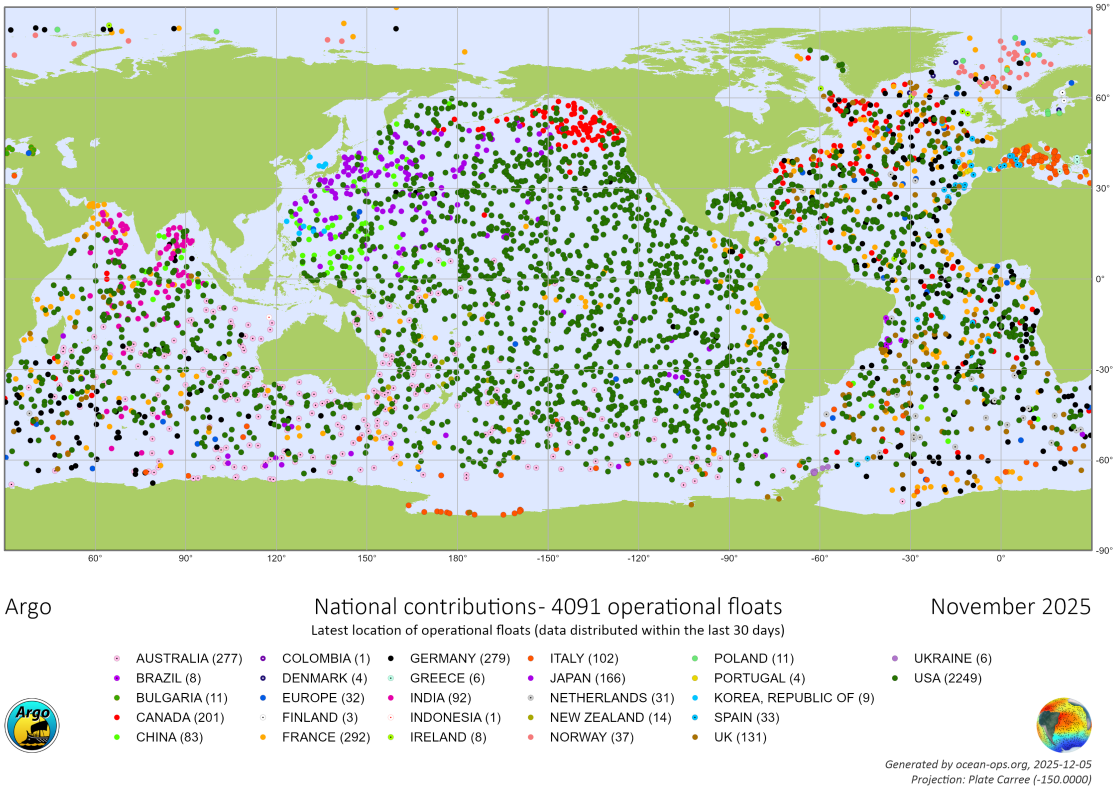


Figure 2: Current Argo Operational Floats (November 2025), Courtesy of Argo Program.

1.1 Problem Analysis

There are numerous problems with buoyancy engines and their application in underwater autonomous vehicles. These problems fall into several different types.

1.1.1 Economical

In 2018, a typical underwater glider operating on a buoyancy engine would cost between 50000 and 250000 USD (Williams), a price that has not changed as of 2025. Most of these expensive underwater gliders are closed-source, making it impossible to replicate and customize. Additionally, many Argo floats are never picked up or maintained after deployment due to the high costs of deploying and recovering these floats. Retrieving

these devices would also require the use of ships or crafts that would negatively affect the environment. These floats cost between 20000 and 185000 USD to produce.

There is an unfulfilled demand for non-complex and inexpensive, economical buoyancy engine concepts that are capable of precise depth control and can be built and tested at a small scale.

1.1.2 Path Planning

A large part of an underwater vehicle's navigation system is path planning. Path planning is present in any autonomous robot, but compared to land robots, underwater vehicles must endure the harsh environment of the ocean with certain conditions, like currents and water pressure. The path planning algorithms are of four different types: methods based on geometric model searches, probabilistic sampling-based algorithms, artificial potential field algorithms, and intelligence algorithms. Even with this many path-searching algorithms, a lot of them lack the ability to adapt to the oceanic environment, and, additionally, they perform at a slow speed. The unpredictable movements and variations of the ocean make path planning difficult to develop and follow (Cai et al.(Li et al.).

1.1.3 Energy Consumption

Current floats and profiling AUVs predominantly rely on their battery capacity, which may sometimes lead to constraints and certain limitations to the operational life of these devices. The buoyancy engine is the largest energy consumer in an AUV, and this consumption only increases with depth, since the buoyancy engine needs more power to function under high pressures, leaving the remainder of the energy for the other parts (sensors and microcontroller) (Yu et al.).

1.1.4 Environmental Impact

Argo floats, at the end of their operational life, remain at the sea surface and may impact the marine traffic. Additionally, they slowly erode over a period of time and sink to the seabed, possibly causing issues for the ocean's wildlife and contributing to the pollution of the seas. The main substances released include aluminum, copper, zinc, lithium, lead, plastic, and tributyl tin oxide (TBTO). Although these quantities are a lot smaller than the natural fluxes, they should still be taken into account. Additionally, there is also the risk of floats and other devices being pushed by the currents into protected or restricted areas, causing possible harm to the fauna and flora of these areas.

1.2 Problem Statement

Considerations: Some of the problems stated in the problem analysis (1.1) are not addressed in the problem statement, since they do not fit this project's scope, and also due to a lack of experience and numerous other constraints, which are stated in 3.1.2, but they should still be analyzed and mentioned in this report.

Conventional buoyancy engines are often too costly and complex for academic or small-scale research applications. This raises the question: **In what possible way can a simple and economical buoyancy engine be built, tested, and improved to demonstrate autonomous vertical movement?** This statement concludes the scope of our project and establishes the basic principles and foundations it relies on.

Chapter 2

Background and theory

2.1 Literature review

2.1.1 Existing solutions

Conventional buoyancy-driven underwater vehicles, namely, consist of three main types in oceanographic research: profiling floats, underwater gliders, and hybrid VBS (Variable Buoyancy System)-equipped AUVs. They differ in buoyancy-engine mechanism, system, and purpose. Hence, it is valuable to review these solutions as they determine the underwater vehicle's capabilities in observing and analyzing ocean properties.

A profiling float is dropped vertically into the ocean, where it drifts into the depths by controlling the buoyancy of the vehicle with a hydraulic pump, which transfers oil between its external and internal reservoirs. The transfer of oil from the internal reservoir to the external bladder increases the volume of the float while maintaining its mass; thereby, the float becomes less dense and rises. When the hydraulic pump transfers oil from its external bladder to the internal reservoir, the density of the float increases, causing it to sink. A microcontroller determines the amount of hydraulic pump needed to push out oil to control the drift at a particular depth. This is energy efficient, as there are no propellers or thrusters involved in the vertical motion of this vehicle. The compact and streamlined vertical shape of the float minimizes drag during ascent/descent. The Argo missions often operate using the architecture and path represented in Figure 3.

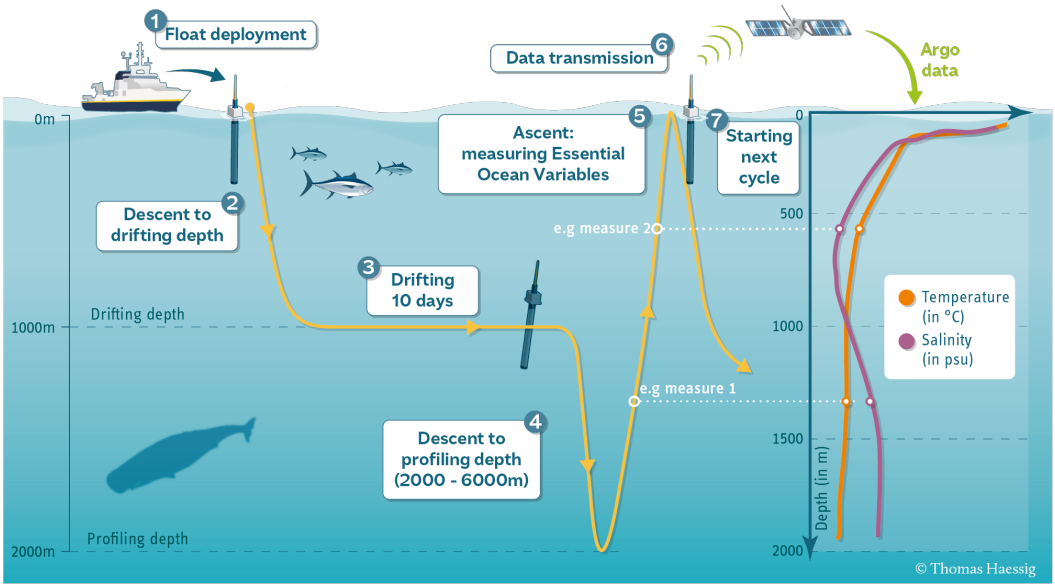


Figure 3: Argo cycle schematic, Courtesy of Argo Program.

More pressure-resistant composite materials in the outer casing complement this trait, in which the Deep Argo mission sinks to a depth of 6000 meters in the ocean. At around 2000 meters of depth, the float begins to profile various oceanic properties such as temperature, salinity, and pressure using a CTD sensor, as well as oxygen level in the water by an oxygen sensor. By sinking to a deeper altitude, the float gains crucial data concerning climate change, since a minor increase in deep water temperature could indicate a huge heat gain in sea warming. Consequently, the profiling float is designed optimally for deep ocean exploration rather than collecting data from a longitudinal plane.

Similarly, underwater gliders apply a water/oil pumping system, but at the front end where the mass is centered, to follow a 'flight path'. A typical glider contains an internal bladder that pulls the back hull upwards on the surface of the water to receive a signal for command. At the front end, it could store several medical syringes or a water tank that pulls water via a buoyancy mechanism to increase density and start the glide. Pumping the water out makes the front end less dense and rise from the front. Hence, a flight path is created where the glider follows a longitudinal and sinusoidal pattern as shown in Figure 4.

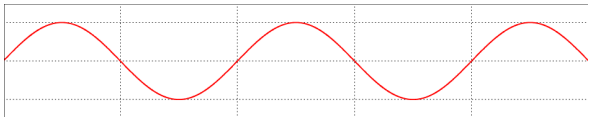


Figure 4: Underwater glider flight path.

A purely buoyancy-driven underwater glider can be energy-efficient in long-term missions up to 8 months, but with a slow speed ranging from 0.15 to 0.30 m/s. Certain variable buoyancy engines could solve this limitation by upgrading motion control with thrusters to increase the velocity. Systems such as pneumatic, water-hydraulic, and thermal are common in VBS:

- A pneumatic system compresses gas to inflate/deflate a bladder and increase the vehicle's volume without adding mass. Hence, it controls the density for vehicle vertical motion control.
- A water-hydraulic system utilizes a reversible pump to draw seawater and change the mass more than its volume to sink or push seawater to lower the density and rise.
- A thermal system uses electrical heaters to warm fluid within a tank to expand it. The fluid then pushes against a flexible bladder to increase the glider's displacement volume to rise or turn off the heaters to cool the fluid and sink.

The choice between the three depends on the objective. For cost and energy efficiency, the water-hydraulic system is significantly simpler and better; the addition of thrusters can relatively improve its agility without affecting the battery consumption too much. Table 2.1 presents a clear comparison between the different buoyancy engine AUV solutions.

2.1.1.1 Strengths and weaknesses of the solutions

Types	Strengths	Weaknesses
Float	- Requires few moving parts - Low maintenance - Cheaper with long operating life	- No horizontal control - Limited sampling frequency in deep depths - Limited payload
Glider	- Can navigate to specific regions in longitudinal plane - Profiles both vertically and horizontally - Decent operating life	- Complexity in mechanical system - Slow velocity - Requires hydrodynamic parts (wings, hull, etc.) and piloting
VBS	- Can operate under confined environments - Flexible in engine types - Space for many sensors	- Complex electronic and mechanical system - Increased payload can affect maneuverability - Expensive

Table 2.1: Strengths and weaknesses of different buoyancy engine-driven AUVs.

The design for this project will be steered towards an underwater vehicle that has a functioning vertical buoyancy control without extra wings. Depending on its efficiency and success, further features (thrusters, sensors, etc.) may be added in order to transition to a VBS device with a water-hydraulic system.

2.2 Theoretical principles

2.2.1 Archimedes' principle

Archimedes' principle states that the object submerged in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced by the object. The buoyant force acts in opposition to gravity.

$$F_b = \rho_{\text{water}} \cdot g \cdot V_{\text{displaced}} \tag{2.1}$$

F_b is the buoyant force,
ρ_{water} is the density of the surrounding water liquid,
g is gravitational acceleration,

$V_{\text{displaced}}$ is the volume of water displaced by the object.

This principle forms the foundation of vertical movement control in the context of a buoyancy engine. By intentionally changing the volume of the engine while keeping the mass constant, the engine directly manipulates the amount of water displaced, and by that changes the magnitude of the buoyant force. An increase in the displaced volume will result in a bigger upward buoyancy force; therefore, the engine will rise. Inversely, a reduction in displaced volume will result in a decrease in the buoyant force, which will allow the engine to sink. Changes in the mass of the device also affect buoyancy; if the total mass increases, it becomes denser than water and the weight force increases, causing it to sink, and if the mass decreases, the density and weight also diminish, causing its ascension.

This makes Archimedes' principle the theoretical core mechanism to control ascent, hovering, and descent in an underwater environment.

2.2.2 Density and Buoyancy balance

The vertical movement of the buoyancy engine is determined by the effective density relative to the surrounding liquid. The effective density can be defined as

$$\rho_{\text{object}} = \frac{m}{V} \tag{2.2}$$

ρ_{object} is the density of the object,
m is the total mass of the object,
V is its external volume.

The engine will rise, sink, or hover according to these conditions:
$\rho_{\text{object}} < \rho_{\text{water}}$ the engine will rise because of positive buoyancy.
$\rho_{\text{object}} = \rho_{\text{water}}$ the engine will hover because it achieved neutral buoyancy.
$\rho_{\text{object}} > \rho_{\text{water}}$ the engine will sink because of negative buoyancy.

In some buoyancy engine configurations, vertical motion is achieved primarily through changes in the total mass of the system rather than changes in external volume. By allowing water to enter or leave the buoyancy engine, the effective mass of the vehicle is changed while the surrounding fluid density remains the same. An increase in the amount of water contained within the system increases the total mass. That results in a higher effective

density than the surrounding water, causing the vehicle to sink. On the other side, extracting water from the system reduces the total mass, lowering the total mass and generating positive buoyancy, and therefore the vehicle will ascend.

2.2.3 Hydrostatic pressure and depth dependency

A buoyancy engine underwater is exposed to increasing external pressure with depth. The pressure is defined by the hydrostatic equation:

$$P = P_{\text{atm}} + \rho_{\text{water}} \cdot g \cdot h \tag{2.3}$$

P_{atm} is atmospheric pressure at the surface
ρ_{water} is the density of water
g is gravitational acceleration
h is the depth below the water surface

The external pressure directly affects the ability of the engine to expand or compress its external volume. The deeper the engine sinks, the greater the external pressure, which will cause the actuators in the engine to exert greater force to change the mass. This causes the energy consumption to grow with increasing depth. Both precision control and energy efficiency depend on the depth.

2.2.4 Energy considerations in volume modulation

Changing the external volume of the engine requires work against the pressure of the surrounding liquid. This work during compression and expansion is defined as:

$$W = \int P dV \tag{2.4}$$

W is the work,
P is the pressure acting on the engine,
dV is the differential change in displaced external volume.

In different cases, to change either external volume or total mass, the engine consumes energy. The amount of energy depends on the depth, which can be calculated through pressure, and the total change in volume. This is critical for energy-efficient engines, but should also be considered in the case of an engine powered by batteries. If the energy consumption of the engine is low, the mission time, or operational time, will increase.

2.2.5 Integrated buoyancy control

The operation of the buoyancy engine comes from the interaction of buoyant force, density balance, hydrostatic pressure, and energy requirements of the external volume or mass manipulation, mentioned in previous subsections 2.2.1, 2.2.2, 2.2.3 and 2.2.4. This subsection will describe the combination of the previously mentioned principles to achieve a buoyancy engine design. The central mechanism is the engine's ability to change its total mass and weight using pistons mounted inside a sealed cylinder with control.

According to Archimedes' principle, the buoyant force acting on the engine is equal to the weight of the displaced water. While the displaced volume of the engine remains constant, vertical movement is achieved by altering the effective mass of the system. By increasing or decreasing the amount of water contained within the engine, the effective density of the system is modified relative to the surrounding fluid. When the pistons are fully extended, water is expelled from the engine, reducing the total mass of the system while maintaining the same displaced volume. This decrease in effective density results in positive buoyancy, causing the engine to rise. Conversely, when the pistons are fully retracted, water is drawn into the engine, increasing the system's mass. The resulting increase in effective density produces negative buoyancy, causing the engine to sink.

Hydrostatic pressure introduces an additional dependency in the buoyancy control process. As depth increases, the surrounding pressure rises, directly influencing the force required to move water into or out of the engine. At greater depths, higher external pressure resists the motion of the pistons, requiring increased actuator force to achieve the same change in internal water mass. This imposes practical limits on the operational depth of the system.

The energetic cost of buoyancy control is therefore linked to the work required to move water against hydrostatic pressure. Both the intake and expulsion of water require mechanical work, and the energy consumption increases with operating depth and the magnitude of mass transfer. For battery-powered systems, minimizing this energy expenditure is critical for extending mission duration.

In summary, buoyancy control in the presented engine is achieved through controlled variation of the system's mass while maintaining a constant displaced volume. The interaction between buoyancy, density, pressure, and energy consumption governs the vertical motion of the engine. Understanding this relationship is essential for achieving stable depth control and efficient underwater operation.

Chapter 3

Design and methodology

3.1 Project Scope and Limitations

3.1.1 Project scope:

The scope of this project is to design, construct, and evaluate an economical prototype of a miniature buoyancy engine that utilizes two longitudinal pistons located at each end of the cylindrical tube to functionally control the AUV's density. The buoyancy engine itself functions based on Archimedes' principle, where the water displacement mechanism works as follows: the pistons retract to pull water into the bladder/tank, increasing the overall density and causing the AUV to descend, and extend to expel water, decreasing density and generating positive buoyancy. Hence, the project aims to:

- Develop mechanical design for the AUV, dual-piston system, and waterproofing/sealing.
- Construct a small-scale prototype using readily accessible engineering materials and/or 3D-printed custom parts.
- Design and integrate a basic actuation method for horizontal piston motion with or without a pressure sensor.
- Evaluate the prototype in a controlled test tank for buoyancy control, response, and overall functionality.
- Document design decisions, engineering calculations, testing procedures, and performance of the prototype.
- Operate in confined environments and avoid large-scale ocean deployment, long-duration operation, advanced control algorithms, and hydrodynamic movement.

3.1.2 Project limitations:

Due to constraints on time, budget, and overall experience, the project will aim for a simple buoyancy engine that can actuate and draw in water to descend and push out water to ascend. There are limitations directly influenced by these constraints, such as depth, actuation, time, resources, and scaling. As first-semester students, it is difficult to mitigate these limitations, as there is a lack of knowledge, resources, and expertise in using high-precision machines and high-tolerance components to achieve optimal sealing, piston alignment, and friction reduction. As a result, leakage and mechanical inefficiencies may occur and affect the performance.

Pressure and depth limitations: The prototype will be designed only for shallow-water testing in a controlled environment, such as a test pool. The expected Blue Robotics acrylic cast tube has a depth rating of 100 meters due to material strength, tube wall thickness, and seal integrity, which are not suitable for high-pressure and deep-water environments. Lack of options occurred as there were limited available tubes on campus, and a custom transparent tube would be difficult to produce due to a lack of expertise. However, the chosen tube is designed and tested for AUV/ROV use and is among the cheapest Blue Robotics tube designs.

Actuation system limitations: The actuation method used for piston movement is simplified and not optimized for efficiency, speed, or long-term reliability. If the pressure sensor is able to detect and follow the program without major errors, the actuation system could be automated to a certain extent.

Testing environment limitations: Full testing will be conducted in a small, static water tank/pool once the prototype passes dry testing. The prototype may face major difficulties under varying temperatures, currents, and open-water conditions; hence, real-world performance will be limited.

Time and resource constraints: Due to the limited project duration of 3 months and material availability, the design first and foremost prioritizes the core functionality of controlled sinking and rising over advanced features such as hydrodynamic and directional control, CTD sensors, thrusters, or multi-cycle endurance trials.

Scale limitations: The behavior of small-scale prototypes does not always scale linearly to larger systems. Drag, inertia, and structural requirements differ significantly for full-sized implementations such as Argo floats and Slocum gliders.

3.2 System Requirements

Part	Requirement statement	Type	Priority	Verification method
Buoyancy Engine	The buoyancy engine should be able to retract and draw water in and extend to push the water out, hence change the system's density.	Functional	High	Test
Servo	The servo motor should be able to turn at least 120 degrees.	Functional	High	Test
Motor arms	Each motor arm should be able to extend and retract at least 90 mm.	Physical	Medium	Measurement
Sensor	The pressure sensor should be able to input data to the Arduino and thus output action to the servo.	Functional	Medium	Test/Calibration
Arduino	The Arduino should be able to register inputs with or without the pressure sensor.	Functional	High	Test/Inspection
Cylinder	The cylinder should be able to withstand pressure from depths up to 100 m.	Non-functional	Low	Stress test
Prototype weight	The total system should be 2.5 kg.	Non-functional	Medium	Measurement
Flanges	Flanges should be designed to fit 2 O-rings that will waterproof the system.	Physical	Medium	Inspection
Rails	Rails should be fitted inside the cylinder with integrated space for pressure sensor wires and customized to stabilize the inner parts.	Physical	Medium	Inspection
Membrane	Membrane shall act as an extra layer for waterproofing.	Functional	Medium	Test
AA batteries	The batteries should be able to support the motor servo with enough power for at least one hour.	Functional	Low	Test/Calibration

Table 3.1: System requirements for the prototype.

The above table (3.1) specifies the necessary features and functions of the prototype parts. It indicates the type, priority, and verification method to describe and highlight

the testing process. The type section states whether the part is functional, non-functional, or simply physical. Functional parts are required to act in motion or movement, and so the testing will be on the efficiency and capability of their action, whereas non-functional parts do not receive inputs or act in motion and thus do not require efficiency tests. The physical parts are essential for functional parts to act, but cannot create any motion on their own. Each part's priority in the prototype is highlighted to show its significance in developing the prototype, and they will be used in the evaluation. Lastly, the verification method simply states the most relevant type of testing to pass the system requirements.

3.3 Design considerations

3.3.1 Mass and Buoyancy

When designing the prototype, the mass and volume need to be taken into account, since both of these affect the buoyancy of the AUV (see equation 2.2). In the design, a change in the total volume of the system will not occur to change buoyancy; therefore, its mass needs to change, via water intake, in order to achieve buoyancy control. The extra water mass that is taken into the AUV, needed for vertical movement, varies when either the volume or the weight of the AUV changes. Since the acrylic tube is 3.078 L in volume, the water mass displaced by the cylinder is 3.078 kg (Archimedes Principle, 2.2.1), so the additional water mass needed to counteract the upward buoyant force is:

$$m_{\text{water}} = 3.078 \; - \; m_{\text{auv}} \tag{3.1}$$

m_{water} is the additional water mass,
m_{auv} is the AUV's initial mass,

AUV's initial mass (kg)	Extra water (kg)	Buoyant Force (N)	Initial Weight (N)
0.500	2.578	30.2	4.91
1.000	2.078	30.2	9.81
1.500	1.578	30.2	14.71
2.000	1.078	30.2	19.62
2.500	0.578	30.2	24.53
3.078	0.000	30.2	30.2

Table 3.2: Additional water mass, in kilograms, needed for different initial cylinder masses to reach neutral buoyancy, and acting forces for each case (the weight refers to the AUV's initial mass, without the extra water).

After analysis of table 3.2, the ideal mass of the AUV is located somewhere between 2.50 and 3.00 kg, since the buoyant force, acting upwards, is 30.2 N, according to Archimedes'

Principle (stated in section 2.1). The initial mass of the AUV, with all its essential components (all listed in subsection 3.5), is approximately 1.5 kilograms, so weights need to be added to the system. Due to the layout of the design, the total inner volume of the submarine is reduced by around 23.3% because the total length of the cylinder is 300 mm and the inner diameter is 101.6 mm. The electronics take approximately 70 mm of the length and occupy the total diameter of the submarine. The device is then left with only 230 mm of length available for the water intake, which means there are 1.865 liters of volume left inside the cylinder, available for the pistons to work and bring water in (0.933 on each side). The thickness of the pistons also needs to be taken into consideration, which then removes some of the total available length. Since the submarine has 2.5 kilograms of mass (with the weights added), it will only need 0.578 liters of additional water to reach neutral buoyancy, and the additional available volume for water intake should serve to sink the submarine and for better buoyancy control.

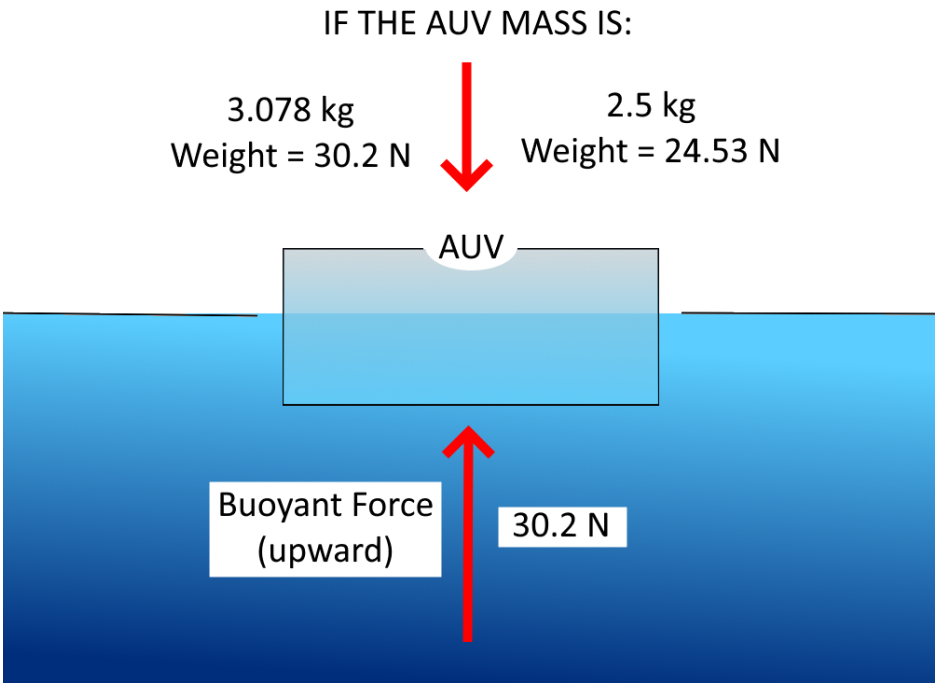


Figure 5: Visualization of the two main forces acting on the AUV, weight and the buoyant force, and the different weight forces for certain masses.

Figure 5 clarifies any confusion regarding the forces acting on the AUV and makes clear that if the device were to have a mass just over 3.078 kg, the weight, the force counteracting the buoyant force, would be greater than it, resulting in the submarine sinking in the water (calculation based on equation 3.2). On the other hand, if it were to have a mass of, for example, 2.5 kg, the weight would be less than the buoyant force. There is no change in

the upward force, since for that to happen, there needs to be a volume change, which will not occur in this prototype.

$$W = m \cdot g \tag{3.2}$$

W is the Weight force,
m is the mass of an object,
g is gravitational acceleration,

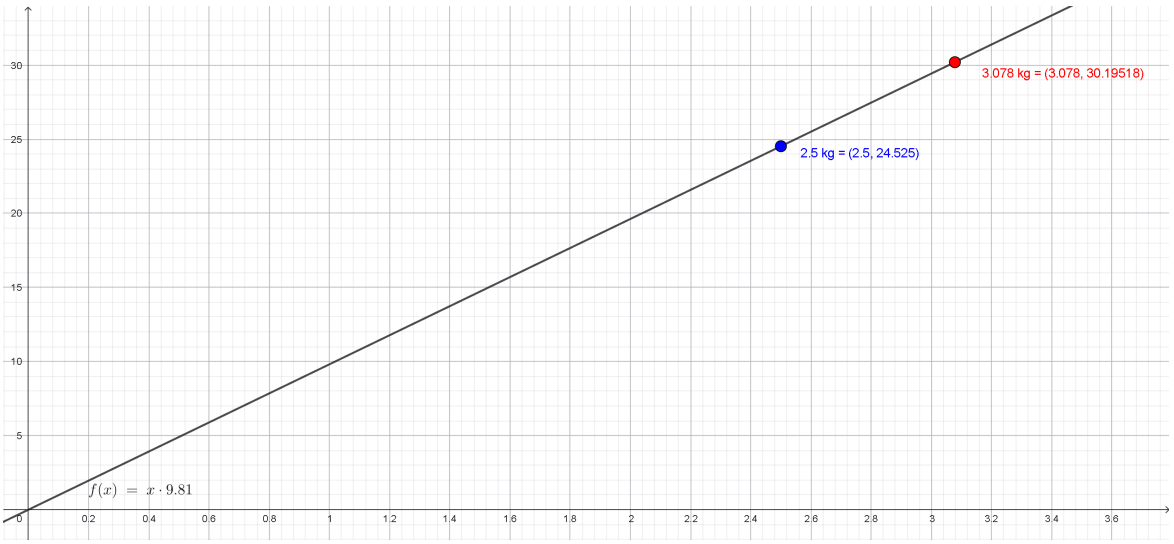


Figure 6: Weight force in relation to different AUV masses, where x is the mass and y is the Weight.

3.3.2 Sealing

The AUV design should be properly water sealed, since inside the tube there are electrical components that will be damaged beyond repair if they were to be in contact with water. For sealing, O-rings should be used in all parts that have a risk of allowing water from the outside environment to infiltrate the system. Additionally, these parts need to have grooves for the O-rings to be placed in effectively.

3.4 Design

At the start of the project and report, a design was thought of, but throughout the semester, some changes were made to account for problems and issues, and so, the final design is a culmination of every member's ideas and solutions.

3.4.1 Initial Design

The initial design, which was planned to work via a motor, that would work similarly to a linear servo (explained briefly in paragraph 3.5) and regular medical-grade syringes that would pull water in and, therefore, change the total mass of the system, had one major problem: the syringes were too long, and could not both fit inside the cylinder and get pulled at the same time by the same motor. The solution to this problem was to make the syringes and pistons ourselves and fit them inside the tube, and seal it. This solution came with issues as well. There was no margin of error; the pistons could not bend or wobble slightly, since if they did, it would cause water to infiltrate into the electronics. There was also an issue with placing the water pressure sensor outside the device without drilling a hole into the tube and without compromising the sealing. The air pressure present inside the tube, after pulling the pistons, could also cause problems for the internal components and cause strain on the pistons, possibly moving them in an unwanted way. The internal pressure would also actively act against the movement of the pistons and harm the motor and the system in charge of moving the pistons by essentially pulling on the rods attached to the motor.

3.4.2 Final Design

The final design has minor tweaks from the initial idea, fixing the most urgent problems. The final design is based around the movement of pistons (3.5-Pistons), facilitated by a positional servo (3.5-Servo) that moves two sets of operating arms (3.5-Operating Arms) connected to the pistons, which are responsible for pulling water into the tube (3.5-Cylinder) and, therefore, changing the device's weight and buoyancy. The design also presents a set of two rails (3.5-Rails) that support the movement of the pistons, removing the instability of the pistons, and allowing for easy communication from the external water pressure sensor (3.5-Sensor) to the internals, since inside one of the rails, there is a path for the cables to pass through. The pistons, in return, also need to be changed to fit these new rails, so they should now have two indents on opposite sides, in order to move along the rails. Membranes (3.5-Membranes) were added to the final prototype to remove the need for near-perfect water sealing in the pistons. This membrane acts as a shield against possible harm to the internal components, caused by water, and as a recipient for the water to stay in, after getting pulled into the device.

3.5 Components selection and list

Before this chapter and its sections, it is relevant to state that the component selection is heavily dependent on their availability on campus and on limitations regarding the acquisition of different parts; there was a set number of suppliers to choose components from. Thankfully, several parts and components of the designed AUV were readily available on

campus, and it was therefore easier to use those than to order them and waste valuable time while waiting for their arrival.

Flanges: Flanges are custom-designed because of the additional required features. Their main purpose is to seal the walls of the cylinder and provide a mounting point for the end caps. Blue Robotics flanges were non-applicable for this design, since there was a need to accommodate a cable port (pressure sensor cable), an external mounting point for the sensor, and an inner mounting point for the internal rail system. The flanges were designed in Autodesk Fusion and afterward resin printed. The decision to print the flanges from resin was based on the structural strength and higher water resistance compared to PLA printing. The design was based on the original Blue Robotics flange in order to ensure proper fitting with the cylinder, o-rings, and the end cap. In order to make the connection between the flange, membrane, and the end cap, there is a layer of Tec 7 sealant in the groove of the flange, previously meant for an o-ring. It is important to mention that both of the flanges originate from the same design, which can be seen in Figure 8, but they are different, since one had to be modified for housing the pressure sensor (Figure 7).

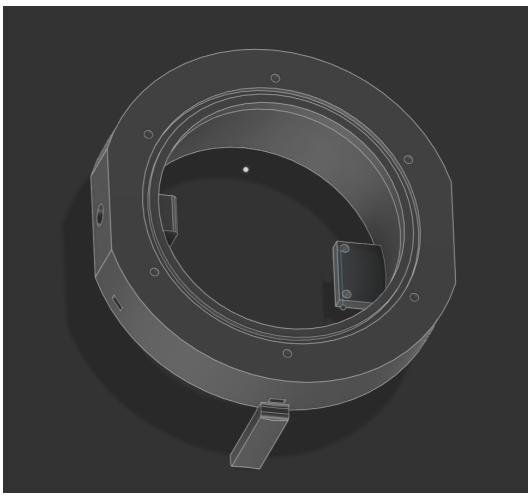


Figure 7: Flange with sensor cable port and external mounting point.

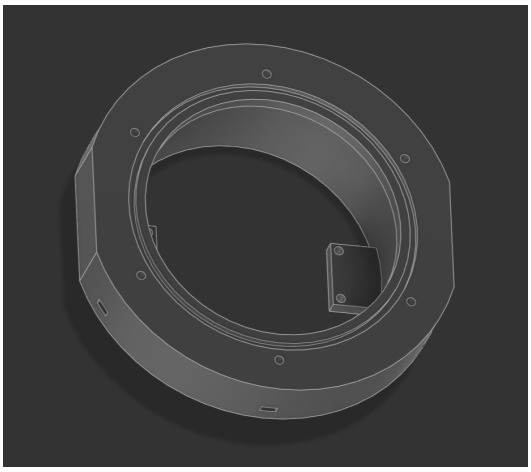


Figure 8: Second flange.

End caps: End caps (see figure 9) primary function is to seal the membrane between the actual end cap and the flange. As mentioned before, the layer of Tec 7 sealant provides a contact surface between our two mentioned components and the membrane to create a waterproof seal. The end caps are mounted using nuts that are housed in the flange and bolts that screw from the top of the end cap to compress the layer of sealant.



Figure 9: Blue Robotics end caps.

Pistons: Two pistons (see figure 10) are either pushing or pulling the membrane inwards or outwards (see figure 11). The membrane is secured using magnets that are glued to the pockets on the surface of the piston. Each piston has a center through-hole that is chamfered on one side to house a screw that connects the piston with the operating arm in order to fit flush with the surface of the piston. The piston mirrors the inner diameter of the cylinder, which has been reduced by 1 millimeter to achieve smooth movement in the cylinder. In addition, there are two guide slots that slot into the inner rail system. Pistons are made from acrylic sheets using a laser cutter.

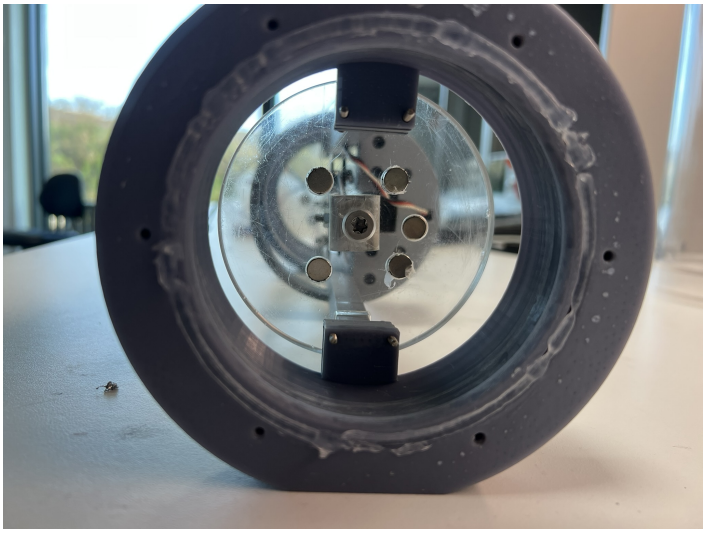


Figure 10: Piston.

Operating Arms: Two operating arms (see figure 11) will transfer the movement of the servo to the pistons and will move said pistons. The arms are two pieces of metal sheet (of different lengths) with two drilled holes at their ends, and they are connected to the servo and the pistons via screws and nuts. There is an additional metal sheet attached to the servo. This piece will connect both arms to the servo. In total, there are 3 metal sheet pieces involved in this operating arm system, with 2 additional pieces, bent into a triangular shape, that are screwed onto the pistons so that the arms have a way to attach to them.

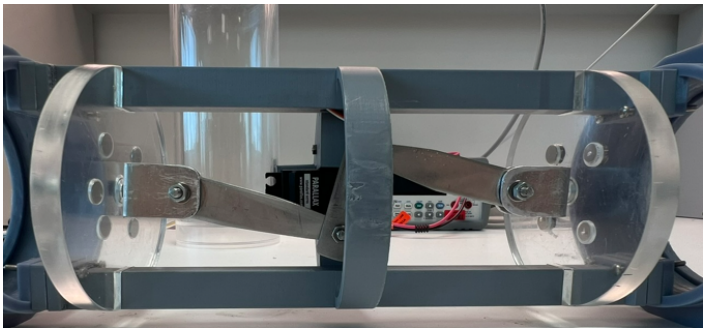


Figure 11: Operating Arms.

Arduino: The microcontroller (see figure 12) board only needs to be able to run a simple program for the buoyancy engine. It needs to be able to obtain the water pressure sensor readings and change the servo position accordingly. Additionally, the Arduino should be small enough to fit inside the tube and not take up too much space, and so, the Arduino

UNO R3 is 68.6 mm in length, that being its longest side, which is within the size limit for the board, stipulated to be anywhere between 0 and 90 mm.



Figure 12: Arduino UNO R3.

Rails: A set of two rails was added to the design of the device (see figure 14); these are attached as part of the flanges and add stability to the motion of the pistons, stopping them from possibly wobbling or dislodging inside the tube, possibly causing water to infiltrate the electronics and destroying them. One of the rails has an opening and an interior tube (see figure 13) that allows the cables from the water pressure sensor to come from outside the submarine and into the internal components, connecting to the processor and any other electronics. Although these rails run along the full length of the AUV, they are not the full length of the cylinder by themselves; rather, they are split into two pieces and, when assembled, are combined. These rails, additionally, contain a support in the middle for the device's internals, which will make the assembly of the electronics much easier.

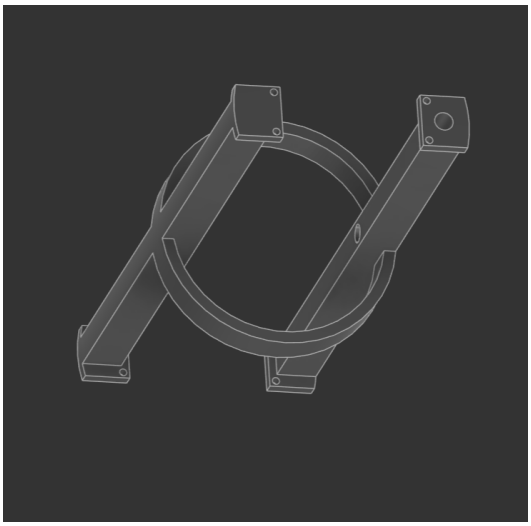


Figure 13: Rails with sensor opening.

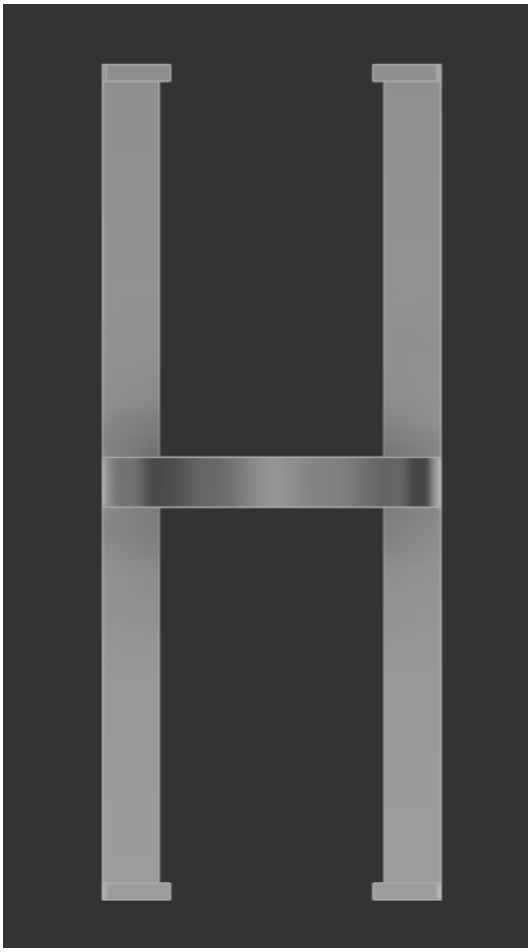


Figure 14: Rails from side view.

Membranes: The addition of two flexible membranes to each piston resolves the issue of water sealing in the pistons, since the membranes will be tasked with holding the water, instead of that issue being handled solely by the piston's water sealing, which, in turn, might be faulty and untrustworthy under the high internal pressure and its own movement. Although the membranes address the issue, they can also rupture or tear, causing internal damage to the electronic components. A tear in the membrane is unlikely to occur, and this project primarily aims to demonstrate buoyancy control at least once. If any critical technical problems related to water damage were to happen after this display, the goals stipulated for this report and project would still have been met.

Servo: The servo should be of the positional type, meaning it is controlled by changing the angle to which it rotates (typically limited to 180 degrees). Continuous servos are controlled by changing the speed at which they rotate continuously, similar to a motor.

Linear servos control straight-line movement (back and forth), as opposed to most other servos that rotate. All these types of servos can change directions, but a buoyancy engine should either use a positional or a linear servo since it needs it to stop and stay at a certain point so that the arms, in the positional servo's case, that pull the pistons don't over-rotate and reach the opposite piston and, possibly, cause harm to the device and its components. The servo chosen was the Parallax Standard Servo (#900-00005) (see figure 15).



Figure 15: Parallax Standard Servo (#900-00005). Courtesy of Parallax Inc.

Servo mount: In order to secure the servo in a fixed position, there is a servo mount (see figure 16 and 17) placed in the center of the cylinder fixed against the rail system. The mount has a pocket where the servo sits tightly and three holes for bolts that secure the mount in place.

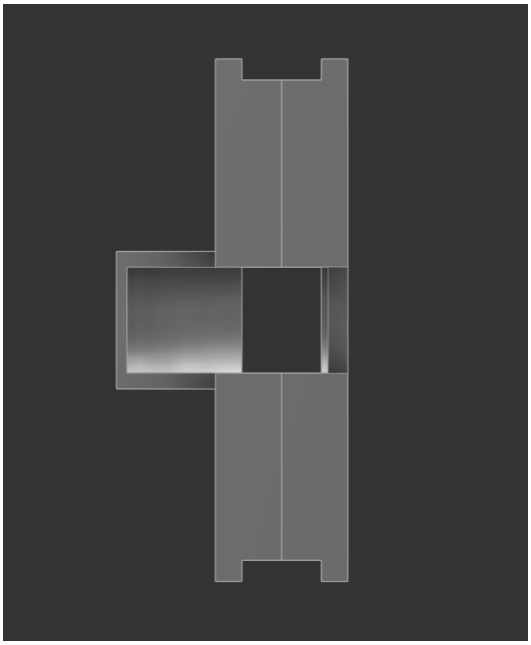


Figure 16: Servo mount top view.

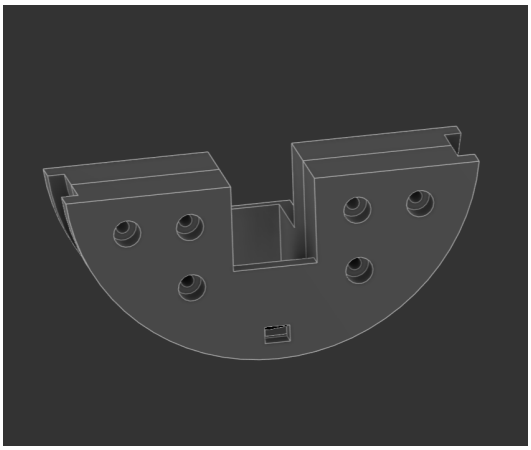


Figure 17: Servo mount.

Cylinder: The tube, which will serve as the shell of the AUV (see figure 18), should not be too large so that the design can accomplish the size goals stipulated for this project, and it should be resistant to relatively high water pressures. It should also have a diameter and length that would allow all the components to fit inside the cylinder and would also have enough space to take water in to change the buoyancy of the system. For all these reasons, the cylinder chosen was the cast acrylic tube, which is 300 mm in length by 114.3 ± 0.8 mm outer diameter with a 100 m water diving limit (BR-101052-300) from Blue Robotics.

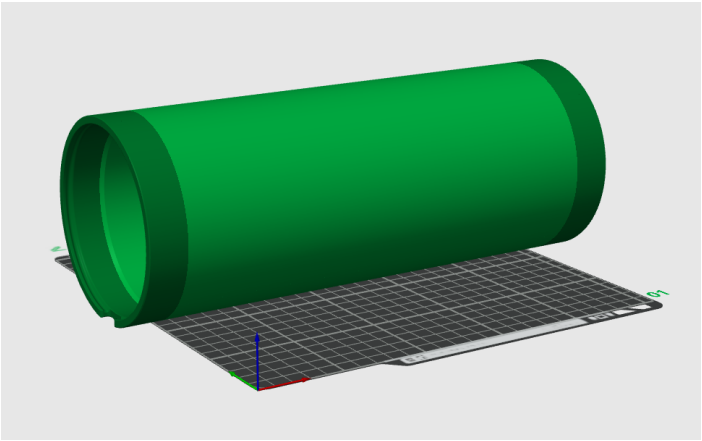


Figure 18: 3D model of BR-101052-300, Appendix (.1). Courtesy of Blue Robotics.

Water Pressure Sensor: A water pressure sensor (see figure 20) will be used to read the depth and regulate the buoyancy of the system accordingly by communicating with the microcontroller board how deep in water the AUV is. The sensor chosen was the Analog Water Pressure Sensor, SEN0257 by DFRobot; it has a measurement range of 0 to 1.6 MPa, which means it can measure up to 163 meters of depth in water. It has a direct correlation between output voltage and pressure (Figure 19) that allows for real-time depth calculations (the code for these is in subsection 4.1.2.2), which, in turn, facilitates autonomy in the AUV's buoyancy control.

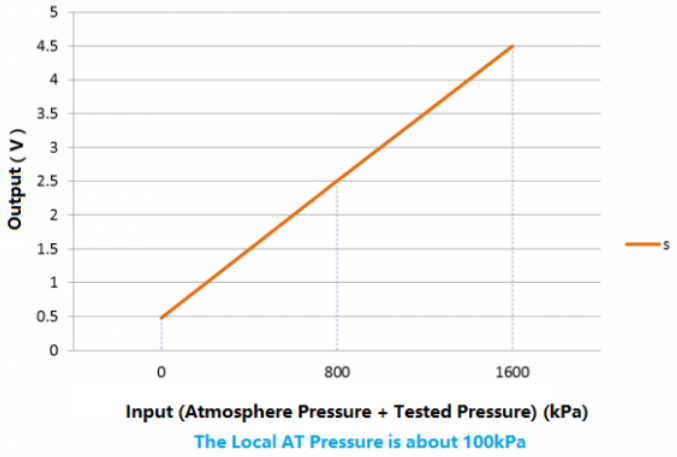


Figure 19: Relation between Voltage and Pressure. DFRobot Gravity: Analog Water Pressure Sensor (SEN0257). Image © DFRobot; source: DFRobot Wiki page for SEN0257 (accessed 11-Dec-2025). Courtesy of DFRobot.

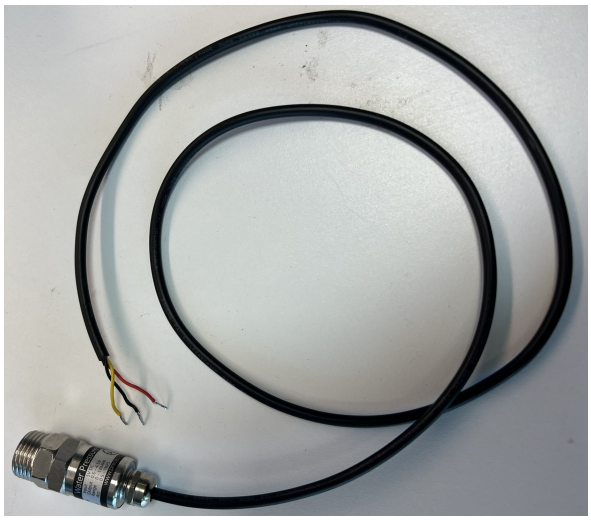


Figure 20: DFRobot Gravity SEN0257 pressure sensor.

3.5.1 Components Listed

Parts/components	Specifications
Flanges and End caps	Customized flanges based on: BR-100665 from Blue Robotics End caps from Blue Robotics BR-102993-005
Operating Arms	Main Arm: Stainless steel 90 mm x 12 mm x 1.5 mm Push/Pull Arm: Aluminium 70 mm x 12 mm x 1.5 mm
Pistons	Acrylic cylinders with m6 cutouts for magnets Diameter: 99.9 mm Thickness: 10 mm
Arduino UNO R3	Main Microcontroller: ATMega328P (16 MHz) I/O Voltage: 5 V USB Connector: USB-B Weight: 25 g Width: 53.4 mm Length: 68.6 mm
Rails	Fully customized (13)
Membrane	Plastic bags
Servo	Holds angles between 0-180° Power: 4 to 6 VCD Dimensions: 5.58x1.9.4.06 cm Weight: 44 g Torque: 0.268 N-m Communication: Pulse-width 0.75-2.25 ms, 20ms intervals
Cast Acrylic Tube	Length: 300 mm Inner Diameter: 101.6 $\pm$ 2.1 mm Outer Diameter: 114.3 $\pm$ 0.8 mm Weight: 746 g Tube Material/Finish: Cast Acrylic Depth Rating: 100 m
Water pressure sensor	Pressure Measurement Range: 0-1.6 MPa Input Voltage: +5 VDC Output Voltage: 0.5-4.5 V Waterproof Level: IP68 Response Time: <2.0 ms Weight: 113 g

Table 3.3: Components list with each component's specifications.

Chapter 4

Implementation and evaluation

4.1 Assembly and integration

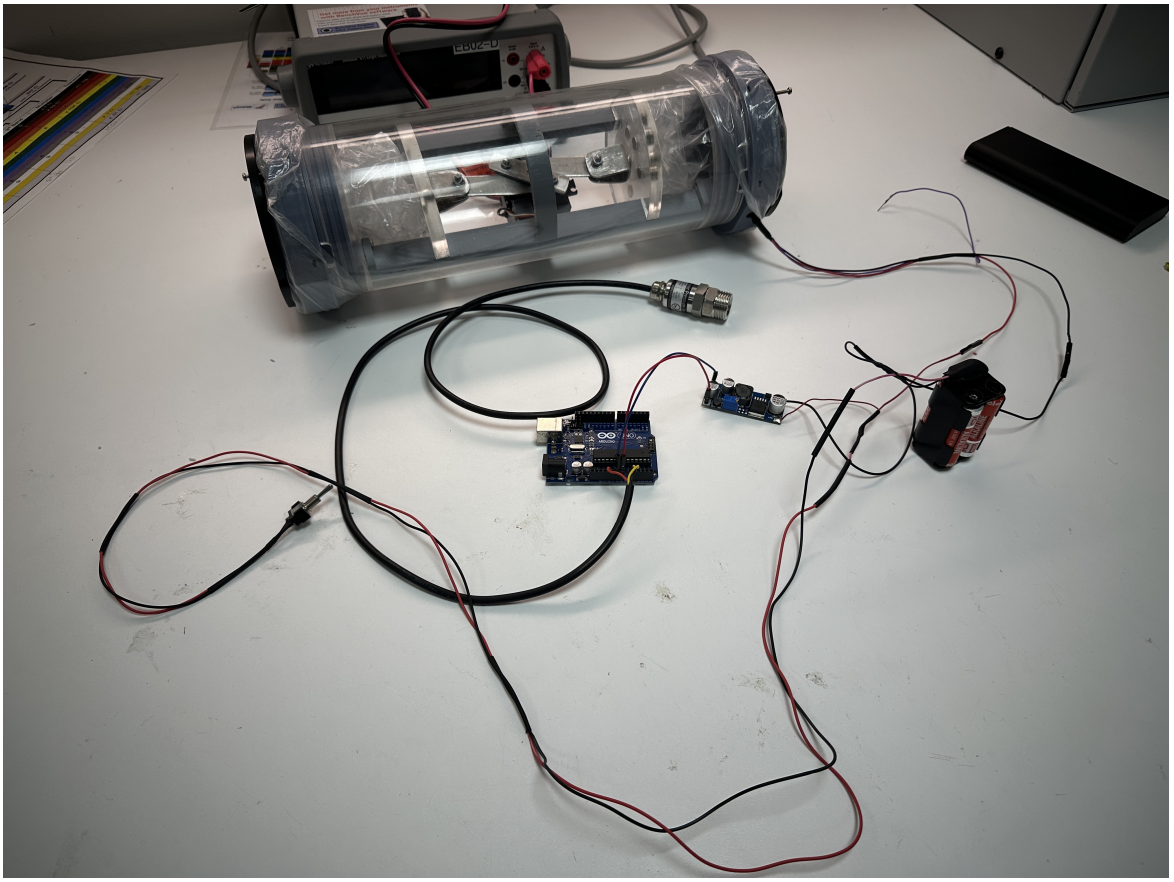


Figure 21: Full view of the prototype.

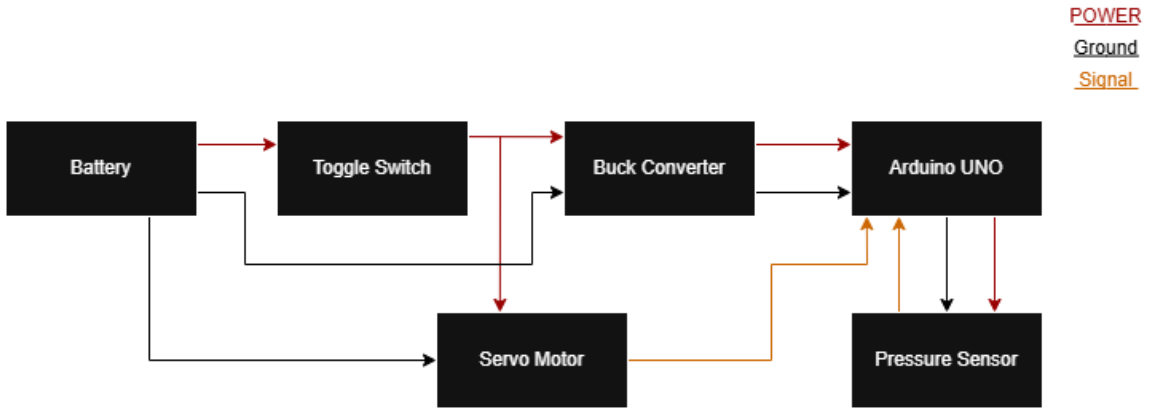


Figure 22: Block diagram.

4.1.1 Hardware

The process of assembly started with the steel arms and the mounting points, followed by the internal frame or rails that held the electronics and pistons in position. Magnets were coupled to the pistons to hold the membrane in place without puncturing it. Then the pistons were placed and aligned onto the rails, and connected to the shaft of the motor. Testing was necessary at this point to ensure that the arms have similar mechanical resistance, as well as that the rails guide the pistons correctly in a constrained linear movement. The battery pack was then placed and fixed onto the frame, while the Arduino and motor, plus the pressure sensor electrical connections, were made and tested. The O-rings were lubricated and installed, and silicone gel was used for the flange top section. The next step was to install the membrane via the magnets and make sure that when the membrane is attached to the flange, it seals watertight. Tightening the screws and placing the enclosures is the final physical adjustment that also includes positioning the pressure sensor correctly for accurate readings.

4.1.2 Software

The full source code is available online in the GitHub repository (see Appendix .5). The software was developed to control the buoyancy engine by collecting and analyzing input from the water pressure sensor, and based on the result, command a servo that is connected to a piston system. The program runs on an Arduino UNO R3 microcontroller. The program operates in a continuous loop, which results in a real-time response to changes in depth.

The main objective of the program is to switch between ascent and sink modes, based on the processed input data, while working between predefined depth thresholds. The

threshold-based approach was chosen to ensure predictable behavior of the system, as well as to simplify debugging and validation of output during early prototype testing.

4.1.2.1 System Overview

The program structure contains three main blocks. Input of pressure sensor data, depth calculation, and piston positioning. The pressure sensor inputs an analog voltage proportional to the water pressure surrounding the engine (see figure 19). The signal is read by the analog input pin on the Arduino board, then the program calculates depth from the pressure value using the hydrostatic equation (see equation 2.3). After the depth is acquired, it's used to determine the appropriate servo position for the buoyancy adjustment. The schematic overview of the login is shown in a flowchart (see figure 23).

4.1.2.2 Sensor reading and depth calculation

The water pressure sensor is connected to an analog input pin of Arduino (see figure 24). Raw sensor input values are continuously sampled and averaged every 2 seconds (see 4.1, line 2).

```
1  void loop() {
2    const unsigned long WINDOW_US = 20000UL;
3    unsigned long start = micros();
4
5    long sumRaw = 0;
6    float sumDepth = 0.0;
7    unsigned long count = 0;
8
9    while (micros() - start < WINDOW_US) {
10     int raw = analogRead(PIN_SENSOR);
11     float v = raw * (5.0 / ADC_MAX);
12
13     float dV = v - offsetV;
14     if (dV < 0) dV = 0;
15
16     float p_pa = dV * P_PER_V_PA;
17     float depth_m = p_pa / (RHO * G);
18
19
20     sumRaw += raw;
21     sumDepth += depth_m;
22     count++;
23   }
```

Listing 4.1: 2 second sampling window

In order to reduce noise and ensure more accurate readings. To account for sensor offset at zero depth, the program sets a static offset in the setup phase (see 4.2, line 8).

```
1  const float P_MAX_MPA   = 1.6;
2  const float P_PER_V_MPA = P_MAX_MPA / 4.0;
3  const float P_PER_V_PA  = P_PER_V_MPA * 1e6;
4
5  const float RHO = 1000.0;
6  const float G   = 9.80665;
7
8  const float offsetV = 0.4744;
9
10
11 const float DEPTH_SINK_MIN = 0.005;
12 const float DEPTH_SINK_MAX = 0.090;
13 const float DEPTH_ASCEND   = 0.50;
```

Listing 4.2: Set up variables

The final offset value was acquired from testing of the pressure sensor. This offset value is then subtracted from the measured voltage input (see 4.1, line 13). In case of a negative value, the value is set to zero to avoid irrelevant readings (see 4.1, line 14). The voltage value is then converted to pressure using the sensor's specified pressure-to-voltage ratio (can be seen in figure 19 and in 4.1, line 16). After the pressure is calculated, the program proceeds to calculate depth from the measured pressure value using the hydrostatic equation (see 4.1, line 17), assuming constant water density and gravitational acceleration (see 4.2 lines 5 and 6).

4.1.2.3 Control logic

The control is based on predefined depth thresholds that define ascent and sink zones (see 4.2, lines 11 to 13). When the calculated depth results are within the sinking range, the servo is moved to the sinking position (see 4.3, line 4). Once the depth reaches or exceeds the ascent threshold, the servo is moved to the ascent position (see 4.3, line 12), allowing the system to regain buoyancy.

4.1.2.4 Servo Actuation

The servo motor is responsible for driving the piston system that adjusts the internal volume of the buoyancy engine. Two predefined servo angles define the ascent and sink modes. The angles were selected experimentally to ensure reliable mechanical operation and to avoid collision with the internal components mounted inside the cylinder, while ensuring a safe operating range of the servo motor. The servo position is changed only

when the control state is updated to reduce mechanical wear and unnecessary power consumption.

```
// SINK ZONE
 if (avgDepth >= DEPTH_SINK_MIN && avgDepth <= DEPTH_SINK_MAX) {
   if (currentAngle != SERVO_SINK) {
     currentAngle = SERVO_SINK;
     myServo.write(currentAngle);
     Serial.println("ENTERING␣SINK␣MODE␣(110degs)");
   }
 }
 // ASCEND ZONE
 else if (avgDepth >= DEPTH_ASCEND) {
   if (currentAngle != SERVO_ASCEND) {
     currentAngle = SERVO_ASCEND;
     myServo.write(currentAngle);
     Serial.println("ENTERING␣ASCENT␣MODE␣(11degs)");
   }
 }
```

Listing 4.3: Control Logic.

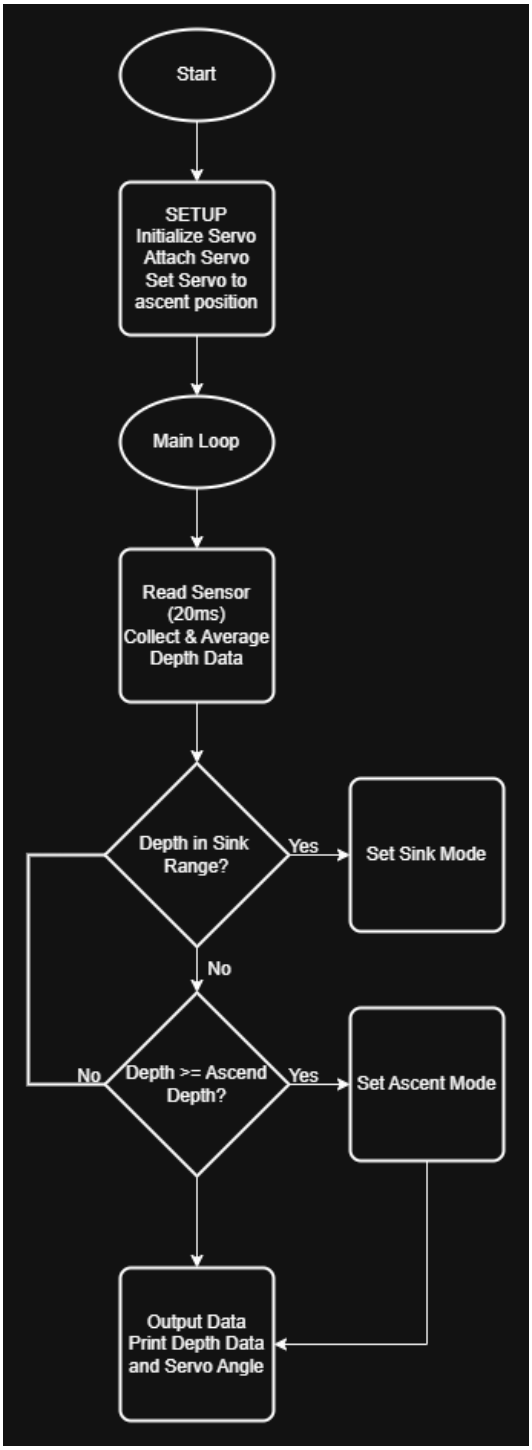


Figure 23: Flowchart of the code.

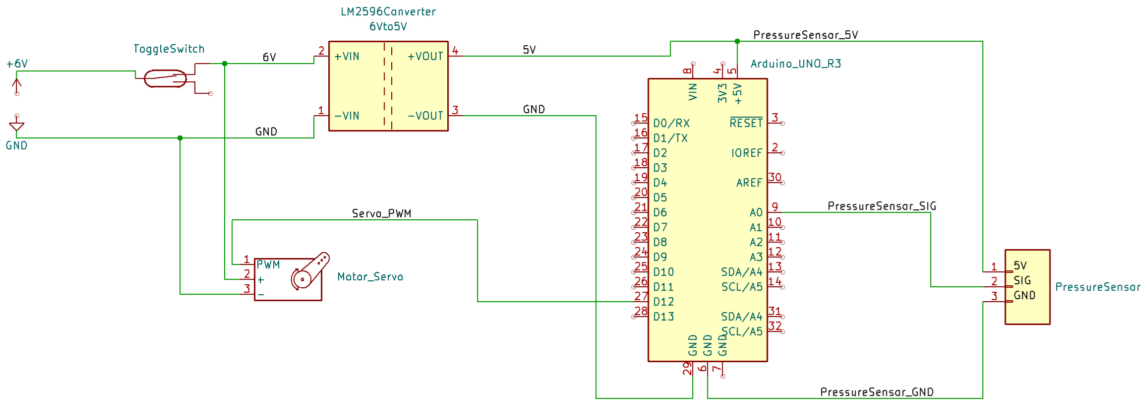


Figure 24: Wiring Diagram.

4.2 Testing procedures

Water pressure sensor To test the pressure sensor, the code written in Arduino IDE was run with the water pressure sensor connected to the microcontroller. Taking the values presented in the serial monitor, inside the program, adjustments were made to the code until the displayed values fit the ones stipulated on the component's datasheet by its manufacturer (.3), and the sensor was working properly. When the depth reading outside the water was accurate, testing inside the water started. The sensor was dipped in water at different depths, and if the values didn't change or were not correct, something was obviously wrong in the code, or the sensor was faulty. The testing was mostly trial and error until the sensor was working as it should and displaying the correct numbers.

Servo The servo works by changing its rotation angle; therefore, to test it, a code was created where an input on the serial monitor would change the servo's angle of rotation. The testing for the servo was, similarly to the sensor, trial and error; if no movement occurred when the servo was given a certain input, changes were made to the code until the servo moved correctly and to the right position.

The next natural step was to connect the servo's rotation to the output given by the sensor. The servo should change its value in relation to the pressure (and depth) detected by the sensor. If the depth is higher than a set value, the servo should rotate to the maximum angle (set in the code) and ascend, but if the depth is lower than or equal to 0, then the servo should rotate to the minimum position and begin its ascension in the water.

4.3 Results and Discussion

The prototype, in its current state, mostly fits the project scope, but the problem statement is yet to be fulfilled. Throughout the project, various challenges arose regarding the design, integration, and space management. All essential parts were designed and acquired within the budget and can fit together comfortably for the assembly. Functionality of the prototype was met with both successes and setbacks. Electronic parts were a success in general, as the coding and wiring process resulted in a functioning motor that could receive input and send output by rotating in a given degree. However, the design of the rails and pistons seems to hinder the arm motion due to the friction caused by the deviated arm expansion. Consequently, full testing was postponed due to the prototype's current state, and these tests would take place soon after January.

In mechanical integration, setbacks occurred regarding the design of the operating arms, which were two long metal arms connected with a central plate on the servo (see Figure 11). The operating arms were carved out of spare aluminum sheets, each 70 mm in length, using a stationary drill. The drilling process was less tedious than expected, as the material is aluminum rather than steel. Afterward, the arms were measured in fully extended length inside the prototype, and a negligible gap was found between the piston and the flange end. For inspection, the pistons were pulled and pushed inside the cylinder manually with hands, and the motor arms demonstrated that they can extend and retract at least 90 mm. Further testing using the motor demonstrated that it could perform the same motion when placed outside the cylinder, but the piston deviation with the rails resulted in unsuccessful cycles. The piston deviation would jam along the rails as the motor arms extended, presumed to be caused by the arms' placement, which extended the piston unequally and sideways. A possible solution would be to redesign the rails and the disks with near-perfect alignment so that no jamming occurs during extension.

The electronic parts, on the other hand, demonstrated optimistic results as the microcontroller board, servo, sensor, and batteries performed their tasks as expected in the requirement statements. The microcontroller board was initially planned as an ESP32 model with access to WiFi and Bluetooth connections. The board was significantly compact and small, which matched our ideal space management. However, the analog input pin of the ESP32 has an upper limit of input voltage at 3.3 V, but the pressure sensor's maximum voltage output is 4.5 V, which is a significantly higher input than the ESP32 pin can handle. Therefore, the change for an Arduino UNO R3 board was made, since it has a range from 0 to 5 V on the analog input pin and can accommodate the pressure sensor's voltage range. To establish a connection between the microcontroller board and the pressure sensor, the three wires were spliced by soldering the conductors together and wrapped in heat-shrink. This created a lengthy, low-resistance joint that could reach the rail sensor opening and allow the pressure sensor to contact outside the prototype. The spliced wires

can be seen in Figure 21. Furthermore, as visualized by Figure 24, the batteries supply 6 V to the buck converter, which steps down the voltage from 6 V to 5 V and then supplies the microcontroller board and the pressure sensor. The servo motor is powered directly by the batteries to achieve the highest possible torque. The connection between the D12 pin on the microcontroller and the servo communicates commands regarding inputs and outputs. The current flow then completes its loop through GND (Ground) wires in the pressure sensor, the microcontroller board, and the converter. Testing without the rails has proven that all electronic processes are functional for turns between 11 and 110 degrees. The block diagram visualizes the simplified version of this process where voltage flow is represented by power lines, current loop path represented by ground lines, and the data flow by signal lines (see Figure 22).

Overall, the total system costs approximately 400 USD, which meets one of the main goals in the project. The cost of 400 USD is a massive price drop for a vertical moving buoyancy engine, as the cheapest commercial AUVs cost around 20000 USD (see 1.1.1). Although the prototype lacks the depth potential and complex profiling sensors that typical AUVs utilize, it certainly could demonstrate promising movement in shallow water, such as lakes and beaches. For idealism, the total system's total mass should be around 2.5 kg to reach and surpass the needed 3.078 kg easily to sink. The current system's mass of 1.5 kg would need an extra 1 kg ballast weight to reach this goal. Regardless, with functioning motor arms that can retract fully, the added water mass would be approximately 1.6 kg, which indicates the total system can reach over 3.078 kg, and thus sink. For further improvements, the prototype could incorporate more resin-printed parts for waterproofing, a CTD sensor for detailed water profiling, a valve closer in the end caps for better water control, and a well-aligned piston. For future works, the problem of the deviated piston will be addressed and fixed, a full demonstration of the system in water will be conducted, and hence, the full evaluation of the ascent/descent testing.

Chapter 5

Conclusion

Buoyancy engine-driven AUVs are essential for oceanic environmental research. They provide a long-lasting device capable of analyzing different biochemical components effectively at extreme depths and sending out this information back, on average, every 10 days. This way, the information on current sea levels is always recent and represents the current status of the environment accurately. These autonomous vehicles, which work based on Archimedes' Principle, have a major issue concerning their pricing; they are excessively expensive, as stated in subsection 1.1.1. Additionally, they are extremely complex, making it difficult to replicate for academic purposes and use (Problem Statement, 1.2); therefore, this report and project take on the task of designing and prototyping a simpler and smaller device capable of performing diving and surfacing movements in water.

Our submarine's vertical movement is designed to work by increasing the system's weight through water intake (see subsection 3.4.2 for the final design). Two pistons on each side of the device are responsible for pumping water inside the tube that acts as the hull of the AUV until the downward and upward forces balance out, reaching what is called neutral buoyancy. In this state, where the forces are matched, the prototype remains still, not moving up or down in the water. If the amount of water inside the tube were to increase, the prototype would sink, and if it were to decrease, it would rise in the water and would surface.

The report presents all the relevant information from extensive research and learning throughout the semester; it analyzes the problems with buoyancy engines and with buoyancy engine-driven AUVs, existing solutions, and different types of AUVs that utilize buoyancy engines, as well as their strengths and weaknesses. Theoretical principles on which the buoyancy engine takes its base were also studied and are presented in this report, as well as minimal calculations to guarantee that the device works properly and as intended. All the components that make up the prototype are also disclosed in this report, seeing their importance in its functioning. The results, which culminate all the work done

throughout this semester, are the last part of this report.

The goal that was set at the start of the semester has not yet been fully fulfilled. However, an economical engine capable of actuating motion to intake/push water was developed, thereby fulfilling the fundamental function of a water-hydraulic float system that initiates ascent and descent in water. Although the prototype does not utilize a CTD sensor that monitors oceanic properties, it achieved the core function of an AUV that can be used for such purposes. Individually, the current work has succeeded in many high-priority parts, such as the servo, Arduino, buoyancy engine, and nearly all medium- to low-priority parts (see Table 3.1). Mechanically, only the jamming of the motor arms is restricting the prototype from actuating, which will be addressed soon.

At the end of this report and project, the group learned various physics concepts, as well as solid knowledge on working with electronic components. The need for a microcontroller and, consequently, the use of the Arduino IDE program forced the less knowledgeable members to learn some coding, and troubleshoot when issues with the water pressure sensor's or servo's code arose.

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Appendix A

Appendix

.1 3D model of Acrylic tube

3D model , from Blue Robotics

.2 Arduino UNO datasheet

Datasheet , from Arduino

.3 SEN0257 Water pressure sensor datasheet

Datasheet, from DFRobot

.4 Parallax Servo datasheet

#900-00005 servo datasheet, from PARALLAX.

.5 Project Repository

GitHub repository with 3D models and project code